

Phasewise Chowla-Free Lag-Two Memory: Exact Max-Plus Optimization and the All-Clock Supremum

RH research program

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Abstract

Fix a finite clock q before taking $N \rightarrow \infty$. We optimize all universally distance-two-safe phasewise lag-two tables

$$\epsilon_n = f_{n \bmod q}(\mu(n-2), \mu(n))$$

whose interpolation coefficient $c_{11}(r)$ vanishes at every phase. This is the phasewise Chowla-free class, not unrestricted memory. Exhausting all 512 local truth tables leaves exactly 192 with $c_{11} = 0$. A subset-dominance reduction first maps them to four relations $0, J, K, I$; only afterward may K be replaced by I , because their full compatibility profiles agree against canonical neighbors. The resulting exact fixed-clock optimum $G(q)$ is a three-state cyclic max-plus problem with arithmetic weights in $\mathbb{Q}\pi^{-2} + \mathbb{Q}\kappa_2$.

At the square clock $q = 36$ this gives the exact strict same-clock gain

$$G(36) = \frac{9}{2\pi^2} - \frac{\kappa_2}{7} > F(36) = \frac{4}{\pi^2}.$$

More generally, for the RH-374 clocks $q_y = 4 \prod_{i \leq y} p_i^2$, an exact run recurrence yields

$$G(q_y) = B_y + \Delta_y, \quad \Delta_y = \mathcal{E}_y \left(\frac{4}{A_y \pi^2} - \frac{\kappa_2}{D_y} \right) > 0, \quad \Delta_y \rightarrow 0,$$

where $\mathcal{E}_y$ counts even-length positive runs. A separate arbitrary-clock argument lifts fixed q to $Q_y = \text{lcm}(q, q_y)$, retains a one-site independent set, and charges every discarded J contribution to $\sum_{p > p_y} p^{-2}$. It proves

$$\sup_{q < \infty} G(q) = B_\infty.$$

No finite-clock attainment or nonattainment statement, growing clock, phasewise $c_{11} \neq 0$ theorem, adaptive-capacity limit, operator, trace, zero model, or implication for the Riemann hypothesis is asserted.

Keywords: Möbius function; squarefree progressions; max-plus optimization; cyclic independent sets; fixed finite clocks.

1 Problem, class, and exact arithmetic

The frozen distance-two capacity allows sign words with no two $+1$ values at distance two. RH-375 identifies the all-clock supremum B_∞ only for universally safe one-site factors [3]. RH-378 classifies the unphased lag-two tables and isolates the coefficient of the ordinary shift-two correlation [4]. Here the clock is phase dependent, but that coefficient is required to vanish separately at each phase.

Definition 1.1 (Phasewise Chowla-free lag-two class). Fix $q \geq 1$. For each $r \in \mathbb{Z}/q\mathbb{Z}$, let

$$f_r : \{-1, 0, 1\}^2 \longrightarrow \{-1, +1\}.$$

On every ternary input $a = (a_n)$, with zero padding before its first site, the output is $\epsilon_n = f_{n \bmod q}(a_{n-2}, a_n)$. The family is *universally safe* if $\epsilon_n = \epsilon_{n+2} = +1$ never occurs for any ternary input. It is *phasewise Chowla-free* if the coefficient $c_{11}(r)$ in

$$zf_r(x, z) = c_{01}(r)z + c_{02}(r)z^2 + c_{11}(r)xz + c_{12}(r)xz^2 + c_{21}(r)x^2z + c_{22}(r)x^2z^2 \quad (1)$$

is zero for every r .

The displayed monomials form a current-zero interpolation basis on $\{-1, 0, 1\}^2$, and the expansion in that basis is unique, as proved in RH-378. The condition is phasewise: cancellation among unknown correlations carrying different phase weights is not assumed.

Let

$$\delta_{q,r} = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{\substack{n \leq N \\ n \equiv r \pmod{q}}} \mu(n)^2, \quad (2)$$

$$\vartheta_{q,r} = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{\substack{n \leq N \\ n \equiv r \pmod{q}}} \mu(n-2)^2 \mu(n)^2. \quad (3)$$

The first two padded sites change neither limit. The progression sieve gives exact local formulae. If $p^a \parallel q$, then

$$\pi^2 \delta_{q,r} = \begin{cases} 0, & a \geq 2 \text{ and } r \equiv 0 \pmod{p^2} \text{ for some } p, \\ \frac{6}{q} \prod_{p|q} (1 - p^{-2})^{-1} \prod_{\substack{p \parallel q \\ p|r}} (1 - p^{-1}), & \text{otherwise.} \end{cases} \quad (4)$$

Put $\kappa_2 = \prod_p (1 - 2/p^2)$. Then

$$\vartheta_{q,r} = \beta_{q,r} \kappa_2, \quad (5)$$

where $\beta_{q,r} = 0$ if $a \geq 2$ and $r \equiv 0$ or $2 \pmod{p^2}$ for some p , or if $2 \parallel q$ and r is even. In every other case,

$$\beta_{q,r} = \frac{1}{q} \prod_{p|q} (1 - 2p^{-2})^{-1} \prod_{\substack{p \parallel q, p \text{ odd} \\ r \equiv 0 \text{ or } 2 \pmod{p}}} (1 - p^{-1}). \quad (6)$$

Both formulae follow by conditioning the squarefree and two-squarefree Euler products on one residue class [2]. In particular,

$$0 \leq \vartheta_{q,r} \leq \delta_{q,r}. \quad (7)$$

1.1 Fixed-clock cancellation

Proposition 1.2 (Phasewise limiting score). *For every fixed finite q and every phasewise Chowla-free family,*

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n \leq N} \mu(n) f_{n \bmod q}(\mu(n-2), \mu(n)) = \sum_{r \bmod q} (c_{02}(r) \delta_{q,r} + c_{22}(r) \vartheta_{q,r}). \quad (8)$$

Proof. Insert (1). The c_{02} and c_{22} terms converge by (2)–(3), and the c_{11} term is absent at every phase. The c_{01} term is a Möbius sum in one fixed arithmetic progression and is $o(N)$ by Davenport cancellation and finite Fourier inversion [1].

For a masked term, use

$$\mu(m)^2 = \sum_{d^2|m} \mu(d).$$

Fix a cutoff R . Combining the phase congruence modulo q with $d^2 \mid n$ or $d^2 \mid n - 2$ produces only finitely many progressions modulo $\text{lcm}(q, d^2)$ for $d \leq R$; Davenport applies to each fixed progression. The absolute tail is

$$\sum_{R < d \leq \sqrt{N+2}} \left(\frac{N}{d^2} + 1 \right) = O(N/R + \sqrt{N}).$$

Thus q and R are fixed first, then $N \rightarrow \infty$, and only afterward $R \rightarrow \infty$. Both c_{12} and c_{21} contributions are $o(N)$. $\square$

The first missing extension is now explicit. If any $c_{11}(r) \neq 0$, the proof meets a phase-weighted shift-two sum

$$\sum_{n \leq N} c_{11}(n \bmod q) \mu(n-2) \mu(n),$$

for which the frozen sources supply no natural-average cancellation. RH-376 identifies even the unweighted $q = 1$ case as the ordinary shift-two Chowla boundary [5].

2 The 512-table census and canonical dominance

For a local table f , write

$$E_f = \{(x, z) \in \{-1, 0, 1\}^2 : f(x, z) = +1\}.$$

Two phase tables E at r and E' at $r + 2$ are compatible precisely when there are no $x, z, w \in \{-1, 0, 1\}$ with $(x, z) \in E$ and $(z, w) \in E'$. Universal safety is exactly this test at every consecutive pair along addition by two.

The exact interpolation census of all $2^9 = 512$ truth tables gives

c_{11}	-1	$-1/2$	0	$1/2$	1
count	32	128	192	128	32

Thus exactly 192 tables enter the declared class. Their nine possible main pairs, multiplicities, and canonical targets are as follows.

c_{02}	c_{22}	count	target	target minus original payoff
-1	0	8	0	δ
-1	1	32	0	$\delta - \vartheta$
-1	2	8	K	$\delta - \vartheta$
0	-1	16	0	ϑ
0	0	64	0	0
0	1	16	K	0
1	-2	8	J	ϑ
1	-1	32	J	0
1	0	8	I	0

Here the four canonical plus-edge relations and weights are

$$\begin{aligned}
E_0 &= \emptyset, & w_r(0) &= 0, \\
E_J &= \{(0, +1)\}, & w_r(J) &= J_r := \delta_{q,r} - \vartheta_{q,r}, \\
E_K &= \{(-1, +1), (+1, +1)\}, & w_r(K) &= K_r := \vartheta_{q,r}, \\
E_I &= \{-1, 0, 1\} \times \{+1\}, & w_r(I) &= I_r := \delta_{q,r}.
\end{aligned} \tag{9}$$

Proposition 2.1 (Canonical subset dominance). *For every one of the 192 tables, the target relation in the preceding table is a subset of its plus-edge relation, and its limiting payoff is at least the original payoff for every $0 \leq \vartheta \leq \delta$.*

Proof. The subset assertion is a finite nine-edge check. The last column gives the payoff difference. Every entry is nonnegative by (7). Replacing a plus-edge set by a subset can only remove possible composable pairs, so the replacement preserves compatibility with both neighbors. $\square$

The compatibility matrix after this reduction is

$$\begin{array}{c|cccc}
 & 0 & J & K & I \\
 \hline
 0 & 1 & 1 & 1 & 1 \\
 J & 1 & 1 & 0 & 0 \\
 K & 1 & 1 & 0 & 0 \\
 I & 1 & 1 & 0 & 0
 \end{array} . \tag{10}$$

In particular, K and I now have identical full incoming and outgoing profiles. Since

$$I_r = J_r + K_r \geq K_r,$$

every K may be replaced by I without changing compatibility. The order of these two operations is essential: $E_K \subset E_I$, so the replacement need not preserve compatibility against an arbitrary noncanonical neighbor. The subset reduction must come first.

3 Exact three-state cyclic max-plus optimization

After the preceding reductions the action set is

$$\mathcal{A} = \{0, J, I\}.$$

Let $C(a, b)$ be the restriction of (10) to this alphabet. Addition by two partitions $\mathbb{Z}/q\mathbb{Z}$ into one cycle when q is odd and two cycles when q is even; for $q = 1, 2$ the components retain their self-loops.

Definition 3.1 (Fixed-clock max-plus value). For an addition-by-two cycle $(r_0, \dots, r_{m-1})$, put

$$\Gamma(r_0, \dots, r_{m-1}) = \max_{\substack{a_0, \dots, a_{m-1} \in \mathcal{A} \\ C(a_i, a_{i+1})=1 \text{ cyclically}}} \sum_{i=0}^{m-1} w_{r_i}(a_i). \tag{11}$$

Define $G^+(q)$ as the sum of (11) over all components.

This is an exact finite optimization, not a numerical scan. For a conditioned start $a_0 = s$, its dynamic program is

$$V_0^{(s)}(a) = \begin{cases} w_{r_0}(s), & a = s, \\ -\infty, & a \neq s, \end{cases} \quad V_i^{(s)}(b) = w_{r_i}(b) + \max_{a: C(a, b)=1} V_{i-1}^{(s)}(a), \tag{12}$$

followed by $\max_{b: C(b, s)=1} V_{m-1}^{(s)}(b)$ and a maximum over the three starts. All weights lie in $\mathbb{Q}\pi^{-2} + \mathbb{Q}\kappa_2$. The artifact stores them as exact pairs and uses a certified rational enclosure of $\pi^2\kappa_2$ for comparisons; an unresolved comparison fails closed.

There is also a useful independent form. Start from the compatible all- J configuration. Replacing J_r by I_r forces the predecessor $r - 2$ to change from J to 0 , for net gain

$$I_r - J_r - J_{r-2} = K_r - J_{r-2}. \tag{13}$$

Two chosen I sites cannot be adjacent on the addition-by-two cycle. Conversely, any zero action not immediately preceding an I may be changed to J : its weight is nonnegative and the J column is compatible with every possible predecessor left by the construction. Thus an optimum is encoded exactly by its independent set of I sites. Consequently

$$G^+(q) = \sum_{r \bmod q} J_r + \text{MWIS}_{+2}(K_r - J_{r-2}), \quad (14)$$

with self-loops retained. The executable certificate checks (12) and (14) independently.

Lemma 3.2 (Input reflection). *If E_r is a universally safe phase family, then*

$$E_r^* = \{(-x, -z) : (x, z) \in E_r\}$$

is universally safe. For $c_{11}(r) = 0$, its main coefficients $(c_{02}(r), c_{22}(r))$ are the negatives of the original pair.

Proof. A composable pair in the reflected family negates to a composable pair in the original family, and conversely. Also $zf_r^*(x, z) = -[zf_r(x, z)]_{(x,z) \mapsto (-x,-z)}$. The even-degree main coefficients therefore change sign, while $c_{11} = 0$ remains zero. The terms outside the main pair vanish by Proposition 1.2. $\square$

Thus the negative of every limiting score is also attainable. We henceforth define

$$G(q) = \max_{(f_r)} \left| \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n \leq N} \mu(n) f_{n \bmod q}(\mu(n-2), \mu(n)) \right| = G^+(q). \quad (15)$$

For orientation, the smallest clocks already show that memory changes the one-site problem:

$$G(1) = G(2) = \frac{6}{\pi^2} - \kappa_2 > 0, \quad (16)$$

whereas $F(1) = F(2) = 0$ because their one-site phase graphs have self-loops.

4 Exact square-clock refinement

Let $3 = p_1 < p_2 < \dots$ be the odd primes and put

$$P_y = \prod_{i \leq y} p_i^2, \quad q_y = 4P_y, \quad A_y = \prod_{i \leq y} (p_i^2 - 1), \quad D_y = \prod_{i \leq y} (p_i^2 - 2). \quad (17)$$

On one period of the RH-374 odd squarefree-support word, let $R_j^{(y)}$ be the number of positive runs of exact length j . All runs have length at most eight, and RH-374 gives their exact Euler-product formula [6]. Retain its odd-run count O_y and define the new symbol

$$O_y = R_1^{(y)} + R_3^{(y)} + R_5^{(y)} + R_7^{(y)}, \quad \mathcal{E}_y = R_2^{(y)} + R_4^{(y)} + R_6^{(y)} + R_8^{(y)}. \quad (18)$$

The notation $\mathcal{E}_y$ counts even-length runs; it is distinct from the Euler products and from the number of sites inside even runs used in RH-374. The locked one-site square-clock value is

$$B_y = \frac{4 + 2O_y/A_y}{\pi^2}, \quad B_y \longrightarrow B_\infty, \quad (19)$$

by RH-374 and RH-375 [6, 3].

Every positive current phase has

$$a_y = \frac{2}{A_y \pi^2}, \quad (20)$$

and every phase at which both current and predecessor supports are positive has

$$b_y = \frac{\kappa_2}{2D_y}. \quad (21)$$

Define $H_y = \kappa_2 A_y / D_y$. The Euler products give

$$H_y = \frac{1}{2} \prod_{i \leq y} \left(1 - \frac{1}{p_i^2}\right) \prod_{\substack{p > p_y \\ p \text{ odd}}} \left(1 - \frac{2}{p^2}\right) \nearrow \frac{1}{2} \prod_{p \text{ odd}} (1 - p^{-2}) = \frac{4}{\pi^2}. \quad (22)$$

Moreover $H_y \geq H_1 = 8\kappa_2/7 > 2/\pi^2$. One elementary lower bound is

$$\kappa_2 = \frac{1}{2} \prod_{p \text{ odd}} (1 - 2p^{-2}) > \frac{1}{2} \left(1 - 2 \sum_{\substack{n \geq 3 \\ n \text{ odd}}} n^{-2}\right) = \frac{3}{2} - \frac{\pi^2}{8} > \frac{1}{4};$$

here $3 < \pi < \sqrt{10}$ suffices. Together with (22), this proves

$$\frac{a_y}{2} < b_y < a_y. \quad (23)$$

Consider a positive odd-support run of length n , bracketed on the left by a zero phase. If M_n is its optimal three-state score, then

$$M_0 = 0, \quad M_1 = a_y, \quad M_n = \max\{M_{n-2} + a_y, M_{n-1} + a_y - b_y\}. \quad (24)$$

The first alternative ends with I and forces the preceding site to zero; the second ends with J . Induction using (23) gives

$$M_{2m} = (m+1)a_y - b_y \quad (m \geq 1), \quad M_{2m+1} = (m+1)a_y \quad (m \geq 0). \quad (25)$$

Thus an odd-length run matches its one-site MWIS score, while each even-length run gains exactly $a_y - b_y$. The odd phase cycle contains two copies of the support word; the even phase cycle adds no such correction.

Theorem 4.1 (Exact square-clock memory formula). *For every $y \geq 1$,*

$$G(q_y) = B_y + \Delta_y, \quad \Delta_y = \mathcal{E}_y \left(\frac{4}{A_y \pi^2} - \frac{\kappa_2}{D_y} \right) > 0. \quad (26)$$

Furthermore $\Delta_y \rightarrow 0$, and hence $G(q_y) \rightarrow B_\infty$.

Proof. Equation (25), summed over the two copies of every run, adds $2\mathcal{E}_y(a_y - b_y)$ to the one-site value B_y , which is exactly (26). RH-374 proves that an exact length-eight run persists, so $\mathcal{E}_y > 0$; inequality (23) proves strictness.

Every even run contains at least two positive sites and distinct runs are disjoint, so $\mathcal{E}_y/A_y \leq 1/2$. By (22),

$$\Delta_y = \frac{\mathcal{E}_y}{A_y} \left(\frac{4}{\pi^2} - H_y \right) \rightarrow 0.$$

Finally $B_y \rightarrow B_\infty$ by RH-374. No monotonicity of Δ_y is used or asserted. $\square$

For $y = 1$, one has $q_1 = 36$, $A_1 = 8$, $D_1 = 7$, $O_1 = 0$, and $\mathcal{E}_1 = 1$. Therefore

$$G(36) = \frac{9}{2\pi^2} - \frac{\kappa_2}{7}, \quad G(36) - F(36) = \frac{1}{2\pi^2} - \frac{\kappa_2}{7}. \quad (27)$$

The sign is exact:

$$\kappa_2 < \prod_{p \in \{2,3,5,7\}} (1 - 2p^{-2}) = \frac{7567}{22050} < \frac{7}{20} < \frac{7}{2\pi^2}, \quad (28)$$

where the last step uses $\pi^2 < 10$. Thus (27) is an exact square-clock strict gain. It is not called the first same-clock gain, because (16) already gives $G(1) > F(1)$.

5 The all-clock supremum

Let $F(q)$ be the RH-375 one-site optimum. Every one-site family embeds in the present class by

$$f_r(x, z) = g_r(z). \quad (29)$$

Its score expansion has $c_{11}(r) = 0$, and its universal safety is unchanged. Hence

$$G(q) \geq F(q) \quad \text{for every } q. \quad (30)$$

The opposite supremum inequality needs a different argument from Theorem 4.1; no same-prime-support saturation law is claimed for memory tables.

Theorem 5.1 (Exact all-clock phasewise-memory supremum). *Over all fixed finite clocks and universally safe phasewise Chowla-free lag-two tables,*

$$\boxed{\sup_{q < \infty} G(q) = B_\infty.} \quad (31)$$

Proof. Fix an arbitrary finite q and an optimal canonical action family for $G(q)$. Choose y so that $p_1, \dots, p_y$ contain every odd prime divisor of q , and put

$$Q_y = \text{lcm}(q, q_y). \quad (32)$$

Lift the q -periodic family to Q_y . The lifted family produces the same output word and the same fixed-clock limit. The clock Q_y is divisible by q_y and has exactly the same prime support as q_y .

On the fine Q_y phases, retain every positive-weight I phase. Retain a J phase only when its predecessor phase is forced divisible by p^2 for some supported prime $p \leq p_y$, including $p = 2$. At such a J phase, $\vartheta = 0$, so its contribution $J = \delta$ is a one-site weight. Delete every zero-density current phase.

The retained positive phases form a distance-two independent set. Indeed, the compatibility column of I in (10) forces its predecessor action to be 0. For a retained J , the predecessor has zero squarefree density by construction. Therefore the total retained weight is at most the one-site optimum at Q_y . The special RH-375 same-support theorem applies only at this one-site step and gives

$$\text{retained contribution} \leq F(Q_y) = B_y. \quad (33)$$

It remains to bound the discarded J phases. Their payoff $J = \delta - \vartheta$ is the density of current-squarefree inputs for which the predecessor is not squarefree. Since the predecessor fine phase was not forced divisible by any supported prime square, every such input has $p^2 \mid n - 2$ for some $p > p_y$. Summing first over the disjoint fine phases, the number of discarded inputs up to N is at most

$$N \sum_{p > p_y} p^{-2} + \pi(\sqrt{N+2}) + O(1),$$

because only primes $p \leq \sqrt{N+2}$ can occur and each congruence $p^2 \mid n - 2$ has at most $N/p^2 + 1$ solutions; the $O(1)$ term absorbs the two zero-padded initial sites. Division by N and then $N \rightarrow \infty$, with q, y, Q_y fixed, gives

$$\text{discarded contribution} \leq \sum_{p > p_y} \frac{1}{p^2}. \quad (34)$$

The clock q , the index y , and hence Q_y are fixed while the $N \rightarrow \infty$ limits underlying (33) and (34) are taken. Combining them yields

$$G(q) \leq B_y + \sum_{p > p_y} p^{-2}.$$

Only now send $y \rightarrow \infty$. Since $B_y \rightarrow B_\infty$ and the convergent prime-square tail tends to zero, $G(q) \leq B_\infty$ for every fixed q .

Conversely, (30) and the RH-375 theorem give

$$\sup_q G(q) \geq \sup_q F(q) = B_\infty.$$

Together the two inequalities prove (31). $\square$

The theorem does not decide whether some finite clock attains B_∞ . The proof supplies neither attainment nor nonattainment for G , and it never chooses $q = q(N)$.

6 Exact executable certificate

The standard-library artifact exhausts all local tables. It evaluates (4) and (6) with exact rational arithmetic and solves (12) in the Euler-symbol basis. Its deterministic finite fixtures are

q	exact $G(q)$	role
1	$6/\pi^2 - \kappa_2$	self-loop boundary
2	$6/\pi^2 - \kappa_2$	two self-loop components
3	$9/(2\pi^2) - 3\kappa_2/7$	small-clock fixture
4	$4/\pi^2$	small-clock fixture
5	$15/(4\pi^2) - 5\kappa_2/23$	small-clock fixture
6	$5/\pi^2 - 3\kappa_2/7$	small-clock fixture
36	$9/(2\pi^2) - \kappa_2/7$	exact square-clock strict gain
180	$73/(16\pi^2) - 25\kappa_2/161$	composite fixture
900	$73/(16\pi^2) - \kappa_2/7$	square-clock fixture
44100	$1177/(256\pi^2) - 1105\kappa_2/7567$	square-clock fixture

For $q = 1, \dots, 6$ and all four larger clocks, the three-state cyclic DP equals the independent-set reduction (14). The square-clock run counts give

$$\mathcal{E}_1 = 1, \quad \mathcal{E}_2 = 23, \quad \mathcal{E}_3 = 1105,$$

and reproduce the corresponding $q = 36, 900, 44100$ formulas. The artifact also checks $\sum_r \delta_{q,r} = 6/\pi^2$ and $\sum_r \vartheta_{q,r} = \kappa_2$ for twelve distinct local normalization patterns.

Finite decimal intervals in the result ledger are labelled **diagnostic_only**. They reproduce the exact rows but play no role in the cofinal proof or in any asymptotic inference. The full 512-row census serialization has SHA-256

aaae39b0af85b13e7cc75baa7170a29f1ac60355443d7b80f3fee06d4af56121.

7 Limitations, route verdict, and Gates

Route A is **G0**. The fixed-clock phase cancellation, exhaustive canonical reduction, exact cyclic max-plus optimizer, square-clock strict gain, and cofinal union-bound theorem form a standalone exact chain. Route B is **STOP_SCOPED**. The first blocker is the phase-weighted shift-two term displayed after Proposition 1.2; neither RH-376 nor any other locked source cancels it in ordinary averages.

The result is limited in four further ways. First, the exact square-clock formula is not extended to arbitrary same-support multiples. Second, no monotonicity of Δ_y is claimed. Third, the finite-clock attainment question is left open. Fourth, the scalar supremum is not an ordinary limit of the adaptive capacity and uses no infinite or growing-clock selector.

Gates A–E remain false/open. These externally prescribed arithmetic tables do not define a canonical intrinsic dynamical spectral determinant, a time-oriented scattering or unitary completion, a self-adjoint generator with an intrinsic $T \log T$ law, a von-Mangoldt prime-power trace, or a completed-zeta divisor equality. Nothing here identifies Riemann zeros, constructs a Hilbert–Pólya operator, or proves the Riemann hypothesis.

8 Conclusion

The phasewise $c_{11} = 0$ memory problem has the same all-finite-clock supremum as the one-site class, even though memory produces exact strict gains at individual clocks. The mechanism is now separated cleanly: local truth-table dominance gives a three-state optimizer; even square-clock runs create the finite gain; and a retained one-site set plus an explicit prime-square tail prevents that gain from changing the cofinal endpoint. Any larger memory theorem must pay for the phase-weighted D_2 terms rather than infer their cancellation from this finite classification.

Reproducibility and declarations

Data availability. All source code, exact outputs, tests, closed schema, source hashes, and archive-verification files are contained in the RH-379 repository directory. No external dataset was created or analyzed.

Ethics. This mathematical study uses no human participants, animals, personal data, or biological materials. Ethics approval was not required.

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AI-assisted workflow. The manuscript and executable checks were prepared with AI-assisted academic writing and coding tools under the repository’s documented multi-agent workflow. The mathematical claims are bounded by the frozen sources, exact proofs, independent audits, and reproducible artifacts. The authors retain responsibility for the final content and its integrity.

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